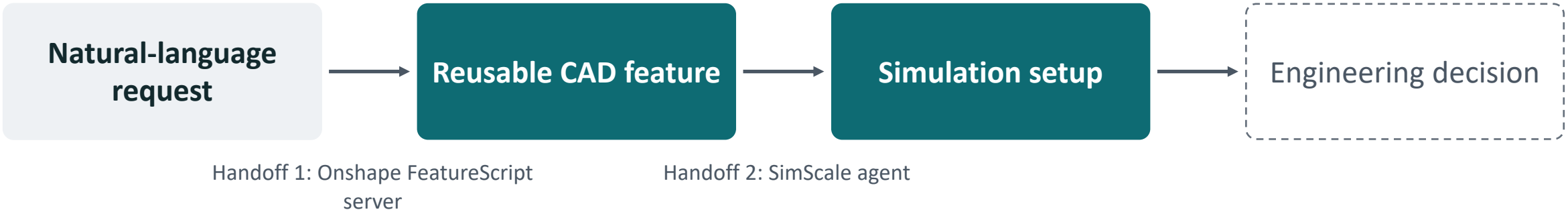
AGENTIC 3D · OCTOBER 16, 2026 · AI-GENERATED NARRATION

AI Enters the CAD-to-Simulation Handoff

For technical product managers and engineering leaders

AI now sits at two engineering handoffs



Workflow announcements, not validation

Evidence labels: REPORTED FACT · SOURCE CLAIM · EDITORIAL INFERENCE · UNVERIFIED · TRANSCRIPT · NOT ESTABLISHED

Source: Onshape, Sept. 30, 2026; Engineering.com, Sept. 23, 2026

Onshape: from a sentence to CAD code

SOURCE CLAIM

Onshape says its FeatureScript MCP Server connects AI language models to Onshape



Code for reusable CAD functionality that adapts to its inputs, not a one-off shape

Text → code → CAD

Requirements travel from the request into Onshape as code a team can read.

EDITORIAL INFERENCE

Easier to inspect: what a team can review

- ☐ Code
- ☐ Parameters
- ☐ Geometry
- ☐ Encoded design rules

Source: Onshape, “5 Ways Teams Can Leverage the Onshape FeatureScript MCP Server,” Sept. 30, 2026

What Onshape's description does not show

SOURCE CLAIM

OBSERVED

Onshape's product description

Supports what Onshape says the server enables.

A valid-looking feature is not, by itself, proof that the design rules are right.

NOT SHOWN

No disclosed test of generated FeatureScripts across:

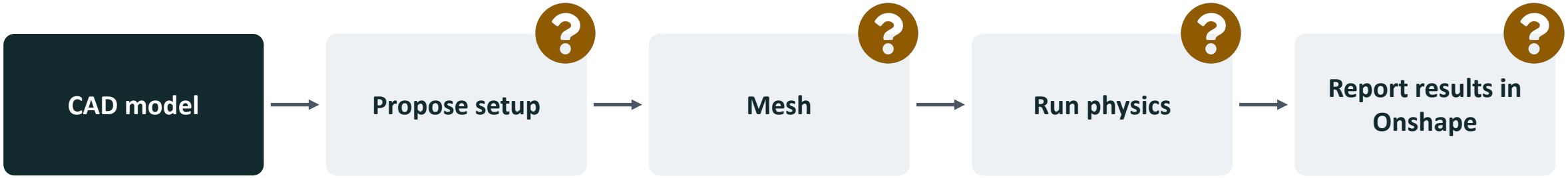
- Requirements
- Edge cases
- Revisions
- Manufacturing outcomes

NOT ESTABLISHED

Source: Onshape, “5 Ways Teams Can Leverage the Onshape FeatureScript MCP Server,” Sept. 30, 2026

SimScale moves the agent downstream

SOURCE CLAIM SimScale says that, from one prompt, its Engineering AI Agent can:



Each simulation step is a claim, not a checked result

Available through Onshape's app store REPORTED FACT

A completed run is not evidence of a valid analysis.

Source: Engineering.com, “Multiphysics gets MCP server and Onshape gets SimScale agent,” Sept. 23, 2026

One prompt, two verification burdens

Reviewing a generated CAD feature

- Code
- Parameter behavior
- Geometry
- Encoded rules

Reviewing an agent's simulation

Does the model answer the engineering question?

- Loads
- Boundary conditions
- Material assumptions
- Contact definitions
- Mesh choices
- Solver settings
- Result interpretation

What the sources checked

REPORTED FACT The integration is available

SOURCE CLAIM What SimScale says its agent can do

NOT ESTABLISHED Not presented: a public comparison with engineer-reviewed setups, or a physical validation study

Source: Engineering.com, “Multiphysics gets MCP server and Onshape gets SimScale agent,” Sept. 23, 2026

Robotics: context, not proof

SOURCE CLAIM

Newton physics for robot simulation

“Completely open source”

A presenter's statement in an NVIDIA video

Supports: NVIDIA's claim about availability and how others can contribute

Does not show: simulator accuracy, transfer to real hardware, or usefulness for a CAD-to-simulation workflow

UNVERIFIED · TRANSCRIPT

A blind robot evaluation, as reported

10
manipulation tasks

3
attempts per task

Shared
time window

Shared
scoring sheet

If verified, closer to useful evaluation discipline.

Automatic transcript: verify against audio, video, and evaluation materials

Source: NVIDIA, Robotics Office Hour, Oct. 15, 2026 (18:10–18:13); Physical AI Podcast, Sept. 26, 2026 (3:36–3:45)

Pilot where review stays explicit

EDITORIAL INFERENCE

1. **Constrain inputs**
   Pilot where inputs are constrained

2. **Review CAD rules**
   Require tests tied to intended rules and reusable parameters

3. **Approve simulation assumptions**
   Reviewers approve the setup before outputs become decisions

4. **Demand public checks**
   Public task suites, disclosed graders, repeatable comparisons with engineer-authored setups, physical validation

What was checked, by whom, and under what conditions?

Source: Editorial recommendation drawing on Onshape (Sept. 30), Engineering.com (Sept. 23), NVIDIA (Oct. 15) and Physical AI Podcast (Sept. 26), 2026